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ORIGINAL ARTICLE

Differences in air displacement plethysmography, bioelectrical impedance analysis and dual-energy X-ray absorptiometry for estimating body composition in Chinese children and adolescents

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Aim: This study aimed to compare body composition (BC) measurements obtained by three widely used BC measuring methods, air displacement plethysmography (ADP), bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA), in Chinese children and adolescents by sex and different BMI categories.

Methods: We used three BC measuring methods to evaluate healthy Chinese children and adolescents aged 5–17 years with BMI categories ranging from underweight to obese. Fat mass (FM, kg), fat mass percentage (FMP, %), fat-free mass (FFM, kg) and appendicular skeletal muscle mass (ASM, kg) were measured by DXA, BIA and ADP on the same day within 1 h.

Results: A total of 172 Chinese children and adolescents were included in this study. The agreements for FM, FFM and ASM estimated by the three methods were excellent or good at the population level (intraclass correlation coefficient > 0.850, $P < 0.05$). However, ADP or BIA estimated lower body fat content and higher FFM than DXA ($P < 0.001$ for all). Moreover, the precise estimates significantly varied across BMI categories. In addition, the limit of agreements was wide, and the differences might not be clinically acceptable at the individual level.

Conclusions: Body fat and FFM obtained by the three commonly used methods were highly correlated, but systematically different and influenced by BMI. This study provided a basis for mutual reference of measurements between three widely used methods.

Key words: air displacement plethysmography; bioelectrical impedance analysis; body composition; children and adolescents; Chinese; dual-energy X-ray absorptiometry.

What is already known on this topic

1 A growing number of guidelines have recommended using body composition analysis methods to accurately assess children's growth and physical development and make targeted clinical decisions.
2 Dual x-ray absorptiometry (DXA), air displacement plethysmography (ADP), and bioelectrical impedance analysis (BIA) are three commonly non-invasive methods, and body composition measurements obtained by the three methods are strongly correlated.
3 The differences between the three methods are inconsistent in various populations. The differences among Chinese children and adolescents, especially under different age groups and body mass index (BMI) categories, are still a gap in research.

What this paper adds

1 It was the first time to compare the differences among DXA, ADP and BIA in Chinese children and adolescents, which provides information about how to interpret body composition measurements by the three methods.
2 The differences of body fat and fat-free mass measured by the three methods was affected by BMI, so more caution should be taken when interpreting the measurements cross BMI category.
3 BIA, a simple, cheap and easy-to-move method, was in good agreement with ADP and DXA at the population level, which provides a basis for the use of BIA in large-scale epidemiological surveys.

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Conflict of interest: None declared.

Author contributions: JM conceived, designed and supervised the study; YH, XW, HD, XS, XZ and XW led data collection; YH, HD and JM analysed data and prepared results; YH wrote the paper with input from all authors; YH, XW, HC, XX and JM revised the paper. All authors finally approved the paper.

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Childhood obesity is on the rise in China and around the world.[1,2] Obesity in childhood tracks into adulthood and is linked to chronic non-communicable diseases, premature mortality, and, in more recent years, the severity of viral infections like COVID-19.[3,4] Body mass index (BMI) is the most common tool used in clinical settings for diagnosing obesity.[5] Nevertheless, BMI fails to distinguish body fat from fat-free mass, whereas measuring the changes in body composition (BC) is essential for evaluating children's growth and development. To overcome the pitfalls of BMI, a growing number of guidelines have recommended combining BC with BMI to accurately assess the physical development of children and make targeted clinical decisions.[6,7]

For BC estimation, multi-compartment models are currently considered the gold standard method. However, this method requires a minimum of three devices to measure different body components.[8,9] Due to its complex procedure and high cost, this method is only used for research purposes. In practice, BC measurements in children and adolescents are estimated more frequently by non-invasive single-device two-compartment (2-C) methods (i.e. air displacement plethysmography (ADP), and bioelectrical impedance analysis (BIA)) and by single-device three-compartment (3-C) methods (i.e. dual-energy X-ray absorptiometry (DXA)). ADP is a relatively rapid method, which makes it easy for children to cooperate. However, its high requirements for clothing and testing environments make it mostly used in research settings. BIA has the advantage of being simple and relatively inexpensive, facilitating clinical use or large-scale epidemiological survey. Both 2-C methods are based on the assumption that the density of fat-free mass (FFM) is constant, but the great variation of water during growth and development will violate this assumption, resulting in inaccurate BC measurements.[8,10] DXA method directly divides the whole body into three components: bone mineral content (BMC), fat mass (FM), and lean mass (LM) through emitting dual-energy X-ray. As it is also recognised as the gold standard method for measuring bone content density, DXA is mostly used in clinical scenarios.[11] However, DXA has small radiation exposure and requires expensive equipment and professional technicians.

The pros and cons of the three BC measuring methods make them suitable for specific scenarios. Several paediatric studies have evaluated differences in measuring body fat content, and FFM of the two 2-C methods (ADP and BIA) compared to the 3-C method (DXA) as the reference method.[12–15] However, these studies were not conducted in Chinese children and adolescents, results were inconsistent, and the differences in different BMI categories were unclear. Therefore, this study aimed to compare BC measurements obtained by three commonly used BC measuring methods in Chinese children and adolescents by sex and different BMI categories.

Methods

Study design and population

In this cross-sectional study, we recruited the convenience sample of Chinese children and adolescents aged 5–17 years from two paths: overweight and obese participants were recruited at the outpatient nutrition clinic in the hospital; underweight and normal weight participants were recruited from the general population by posters in Beijing. All participants reserved their examination date through an online appointment system, and then staff contacted each participant 3 days before the appointment to remind them of the test time and inform them of the preparations and precautions. Participants were excluded from the study if they were pregnant or scoliotic, had limb deformity, had a pacemaker and metallic implants, had weight > 100 kg or height > 200 cm, had severe kidney and digestive diseases, had surgery within 6 months, or had radiographic contrast material within the past 7 days. Each participant was asked not to eat, drink anything, or perform intense physical activities at least 2 h before the examination.

Sample size calculation was based on previous studies comparing BC measurements between ADP and DXA or BIA and DXA. The comparative studies in the past decade were summarised in Tables S2 and S3. These studies reported that the intraclass correlation coefficient (ICC) of two paired methods ranged from 0.37 to 0.99. We used the lowest ICC (0.37) for the sample size calculation to prevent an inadequate sample. The minimum sample size for the present study is 59, with a significance level of 5% ($\alpha = 0.05$) and power at 90% ($\beta = 0.1$). To perform subgroups analysis, we included all participants who met the inclusion and exclusion criteria during the recruitment period. The study protocol was approved by the Institutional Review Board and Ethics Committee of Beijing Children's Hospital of Capital Medical University (Approval no. [2021]-E-155-Y). All participants and their parents provided written informed consent.

Data collection and measurements

All participants underwent a complete physical examination in the same order on the same day within 1 h, with anthropometric measurements first, then DXA scanning, BIA measurement, and finally, ADP test. Before being measured, participants were told to void their bladder, remove all metal items, and change into recommended tight-fit swimsuits. During measurements, participants were instructed to keep quiet, breathe normally, and not move after positioning their bodies as required. We maintained a high level of quality control throughout the study, including well-trained technicians following standardised procedures, daily spine phantom calibration for DXA, daily volume calibration for ADP, fortnightly weight calibration for the weight scale, and maintaining a constant temperature of about 25°C in the test room.

Anthropometry

Weight and height were measured to the nearest 0.1 kg and 0.1 cm using an automatic scaling system (BSM330; Biospace Co., Ltd., Seoul, Korea), with participants barefoot and standing up straight with the head in the Frankfurt plane. All measurements were performed twice, and BMI (kg/m^2) was calculated as the mean value of weight divided by the squared mean value of height. BMI category was classified using age- and sex-specific BMI cut-offs of the International Obesity Task Force (IOTF) criteria, and BMI z-score was calculated using three IOTF-based age- and sex-specific parameters – L, the Box-Cox power; M, the median; and S, the coefficient of variation.[16]

Dual-energy X-ray absorptiometry

Hologic Discovery fan-beam densitometers (Hologic Inc., Bedford, MA, USA) with the APEX software version 4.0. was used for

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whole-body DXA scanning. Participants were asked to lie in the supine position on the examination bed within the designated scan area, keeping their hands flat on the table and their toes pointing slightly inward. The APEX software automatically divided the whole-body weight into three compartments: FM, LM and BMC. Fat mass percentage (FMP) was calculated as FM (kg)/body weight (BW, kg)*100% and FFM (kg) was calculated as LM (kg) plus BMC (kg). Besides, DXA could provide regional BC measurements, and appendicular skeletal muscle mass (ASM) was calculated by adding the muscle mass of the arms and legs.

Bioelectrical impedance analysis

For BIA, we used InBody 720 octa polar multi-frequency equipment (InBody Corp., Seoul, Korea). This equipment has eight electrodes, including two thumb and two palm electrodes in the hand levers, and four foot electrodes on the platform. Before BIA measures, participants were asked to stand still for 5 min and then stand barefoot on the platform with the soles of their feet on the electrodes and grasp the hand levers with their thumbs and figures touching the electrodes. During the BIA testing, participants were instructed to stand comfortably and keep their elbows fully extended with their shoulder joints abducted to a 15° angle.[17] Impedance was measured by passing a small electrical current through the whole body, and then the in-built software calculated overall and regional BC measurements according to a series of equations.

Air displacement plethysmography

For ADP, the BOD POD system (Cosmed Inc., Concord, CA, USA) with software version 5.2.0. was used to measure body volume (BV). Participants were asked to wear a swimming cap that covered all scalp hair and compressed the air. Participants were weighed on the integrated scale, then entered the test chamber for two or three BV measurements. The software predicted thoracic gas volume (TGV) with a validated child-specific equation.[18] BV was converted to body density (BW/BV) with the predicted TGV adjusted. FMP was then calculated using the age- and sex-specific equations of Lohman[19,20] and FM (kg) = FMP (%)*BW, as well as FFM (kg) = BW − FM (kg) were calculated simultaneously.

Statistical analysis

Statistical analysis was performed using R Software (version 4.1.2, www.cran.r-project.org). A two-sided P value < 0.05 was considered to indicate statistical significance. The continuous variables were presented as mean ± SD, and categorical variables were expressed as the frequency (n) and percentage (%). The t test for continuous variables and chi-squared tests for categorical variables were used to compare characteristics between sexes.

Bias was calculated by subtracting ADP-based or BIA-based BC measurements (i.e. FM, FMP, FFM and ASM) from those measured by DXA and BIA-based BC measurements from those obtained by ADP. Paired t tests compared the bias of BC results estimated between paired methods. Agreements were evaluated by computing ICC[21] at the population level and calculating the 95% limits of agreement (bias ± 1.96 SD) using the Bland–Altman method[22] at the individual level. The correlation was classified as excellent (ICC > 0.90), good (ICC: 0.76–0.90), moderate (ICC: 0.50–0.75), and poor (ICC < 0.50).[23] Furthermore, we calculated the mean absolute percent error (MAPE) to rank the performance of ADP and BIA compared to DXA. In the Bland–Altman plots, the difference between the two paired methods is plotted against the mean values of the paired methods.[22,24] Linear regression models were used to examine if BMI z-score contributed to the differences between paired methods' BC results. All results were stratified by sex.

Table 1 Characteristics and body composition of included participants

	Boys (58.1%)	Girls (41.9%)	*P* value†
Age (years), mean ± SD	9.9 ± 3.1	9.5 ± 3.0	0.491
Age group, *n* (%)			0.993
Children	82 (82.0)	59 (81.9)	
Adolescents	18 (18.0)	13 (18.1)	
BAZ, mean ± SD	0.9 ± 1.7	0.6 ± 1.7	0.203
BMI category, *n* (%)			0.385
Underweight	15 (15.0)	15 (20.8)	
Normal weight	40 (40.0)	27 (37.5)	
Overweight	15 (15.0)	15 (20.8)	
Obesity	30 (30.0)	15 (20.8)	
FM (kg), mean ± SD			
ADP measurements	13.1 ± 12.9	9.5 ± 6.9	0.030
BIA measurements	12.8 ± 13.1	9.8 ± 7.8	0.083
DXA measurements	15.3 ± 11.9	12.5 ± 7.2	0.080
FMP (%), mean ± SD			
ADP measurements	24.6 ± 12.7	23.7 ± 8.8	0.604
BIA measurements	23.7 ± 13.6	23.1 ± 10.0	0.746
DXA measurements	30.9 ± 9.2	31.9 ± 6.6	0.403
FFM (kg), mean ± SD			
ADP measurements	31.0 ± 12.6	27.3 ± 10.1	0.040
BIA measurements	31.5 ± 13.0	27.8 ± 10.0	0.048
DXA measurements	29.7 ± 13.5	24.9 ± 9.5	0.011
ASM (kg), mean ± SD			
BIA measurements	12.0 ± 6.4	9.9 ± 4.7	0.020
DXA measurements	12.2 ± 6.0	10.0 ± 4.2	0.008

ADP, air displacement plethysmography; ASM, appendicular skeletal muscle mass; BAZ, BMI-for-age z score; BIA, bioelectrical impedance analysis; DXA, dual-energy X-ray absorptiometry; FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage. † Comparisons between boys and girls were performed using t-test for continuous variables and chi-squared tests for categorical variables.

Results

General characteristics and BC measurements

A total of 172 children (100 boys, 58.1%) aged 5–17 years with valid data of physical examination and BC measured by ADP, BIA and DXA were analysed in the study. Descriptive statistics are presented in Table 1. The mean BMI z-score was 0.9 ± 1.7 in boys and 0.6 ± 1.7 in girls. Boys and girls were evenly distributed regarding age, age group, and BMI categories. Boys had

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Table 2 Comparisons of body composition measured by air displacement plethysmography and dual-energy X-ray absorptiometry between boys and girls

	Boys			Girls			
	ICC	Bias‡	MAPE	ICC	Bias‡	MAPE	*P* for difference†
FM (kg)							
Total	0.971	−2.2 ± 2.1**	27.7%	0.887	−3.1 ± 1.7**	30.7%	0.003
Age group							
Children	0.954	−2.2 ± 1.8**	30.0%	0.838	−3.0 ± 2.0**	31.4%	0.009
Adolescents	0.976	−2.0 ± 3.2*	17.4%	0.927	−3.3 ± 2.0**	27.5%	0.214
P for difference§		0.687	0.021		0.572	0.451	
BMI category							
Underweight	0.726	−2.3 ± 1.1**	44.0%	0.047	−2.5 ± 1.6**	45.4%	0.486
Normal weight	0.659	−2.8 ± 1.5**	37.9%	0.760	−2.9 ± 1.9**	32.2%	0.370
Overweight	0.909	−2.1 ± 1.7**	13.7%	0.722	−3.7 ± 1.5**	25.2%	0.015
Obesity	0.969	−1.5 ± 3.0*	13.0%	0.910	−3.2 ± 1.6**	18.6%	0.049
P for difference¶		0.077	<0.001		<0.001	0.019	
FMP (%)							
Total	0.750	−6.3 ± 5.6**	26.7%	0.501	−8.2 ± 5.1**	29.5%	0.010
Age group							
Children	0.719	−6.9 ± 5.6**	28.9%	0.461	−8.7 ± 5.0**	30.2%	0.059
Adolescents	0.888	−3.2 ± 4.5*	16.6%	0.664	−6.2 ± 5.7*	26.3%	0.112
P for difference§		0.009	0.023		0.111	0.461	
BMI category							
Underweight	0.411	−8.3 ± 4.9**	43.1%	0.022	−9.7 ± 6.4**	44.4%	0.389
Normal weight	0.131	−8.8 ± 4.7**	36.5%	0.280	−8.3 ± 5.9**	31.2%	0.834
Overweight	0.619	−3.8 ± 3.5**	12.6%	0.190	−8.3 ± 3.1**	23.6%	0.001
Obesity	0.487	−3.1 ± 5.9*	12.4%	0.347	−6.6 ± 3.8**	17.3%	0.019
P for difference¶		<0.001	<0.001		0.456	<0.001	
FFM (kg)							
Total	0.981	1.3 ± 2.2**	8.3%	0.959	2.3 ± 1.6**	10.8%	<0.001
Age group							
Children	0.959	1.5 ± 1.8**	8.9%	0.943	2.3 ± 1.6**	11.2%	0.003
Adolescents	0.967	0.6 ± 3.4	5.5%	0.964	2.3 ± 1.9**	9.1%	0.103
P for difference§		0.132	0.019		0.997	0.196	
BMI category							
Underweight	0.953	1.7 ± 1.1**	8.8%	0.870	2.0 ± 1.5**	12.0%	0.305
Normal weight	0.973	2.0 ± 1.4**	9.4%	0.948	2.3 ± 1.8**	11.2%	0.260
Overweight	0.983	0.9 ± 1.9	5.5%	0.954	2.8 ± 1.4**	10.6%	0.013
Obesity	0.975	0.4 ± 3.1	7.9%	0.966	2.3 ± 1.6**	9.1%	0.020
P for difference¶		0.014	0.128		0.004	0.887	

Abbreviations: FM, fat mass; FMP, fat mass percentage; FFM, fat-free mass; ICC, intraclass correlation coefficient; MAPE, mean absolute percentage error. * $P < 0.05$. ** $P < 0.001$. † To test the differences of bias between boys and girls. ‡ Bias was calculated by subtracting ADP-based body composition measurements (i.e. FM, FMP and FFM) from those measured by DXA. The paired *t* test was used to compare the means of bias between ADP and DXA. § To test the differences of bias or MAPE between children and adolescents. ¶ To test the differences of bias or MAPE among BMI category.

significantly higher values of FM measured by ADP and higher FFM measured by all BC methods than girls (all $P < 0.05$).

Comparisons of BC measured by ADP and DXA

At the population level, the agreement between ADP and DXA was excellent or good for FM and FFM measurements (FM: ICC = 0.971 in boys and ICC = 0.888 in girls; FFM: ICC = 0.981 in boys and ICC = 0.959 in girls), but moderate for FMP measurement (ICC = 0.752 in boys and ICC = 0.500 in girls, Table 2). Body fat content (kg for FM and % for FMP) obtained by ADP was significantly lower than that obtained by DXA; however, FFM obtained by ADP was significantly higher than by DXA (bias = 1.3 kg in boys and bias = 2.3 kg in girls, both $P < 0.001$, Table 2 and Fig. 1). At the individual level, The Bland–Altman plots were used to determine the limits of agreement (LOA) between ADP and DXA in boys and girls (Fig. 1). The LOA for FM, FMP, and FFM between the two methods were −6.3 to

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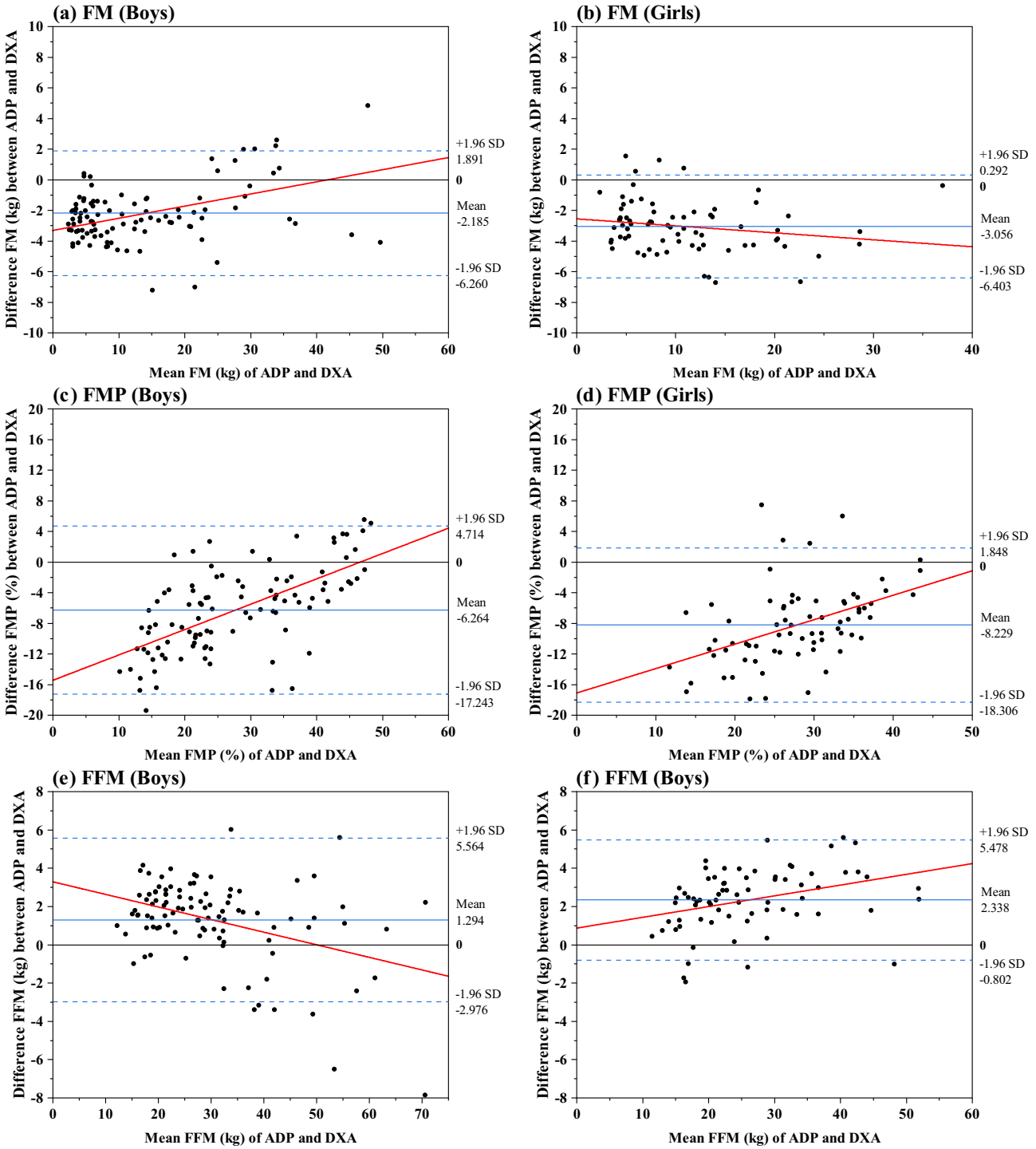

Fig. 1 Bland–Altman plots for body composition measurements assessed by air displacement plethysmography (ADP) and dual X-ray absorptiometry (DXA) among boys (a, c and e) and girls (b, d and f). *Y*-axis represents difference of body composition measurements (i.e. FM, FMP and FFM) measured by ADP minus those measured by DXA. *X*-axis represents mean of body composition measurements (i.e. FM, FMP and FFM) measured by ADP and DXA. Blue solid lines represent the mean difference between measurements. Blue dash lines represent the 95% limits of agreement. Red solid lines represent the linear regression between bias and means of measurements. FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage.

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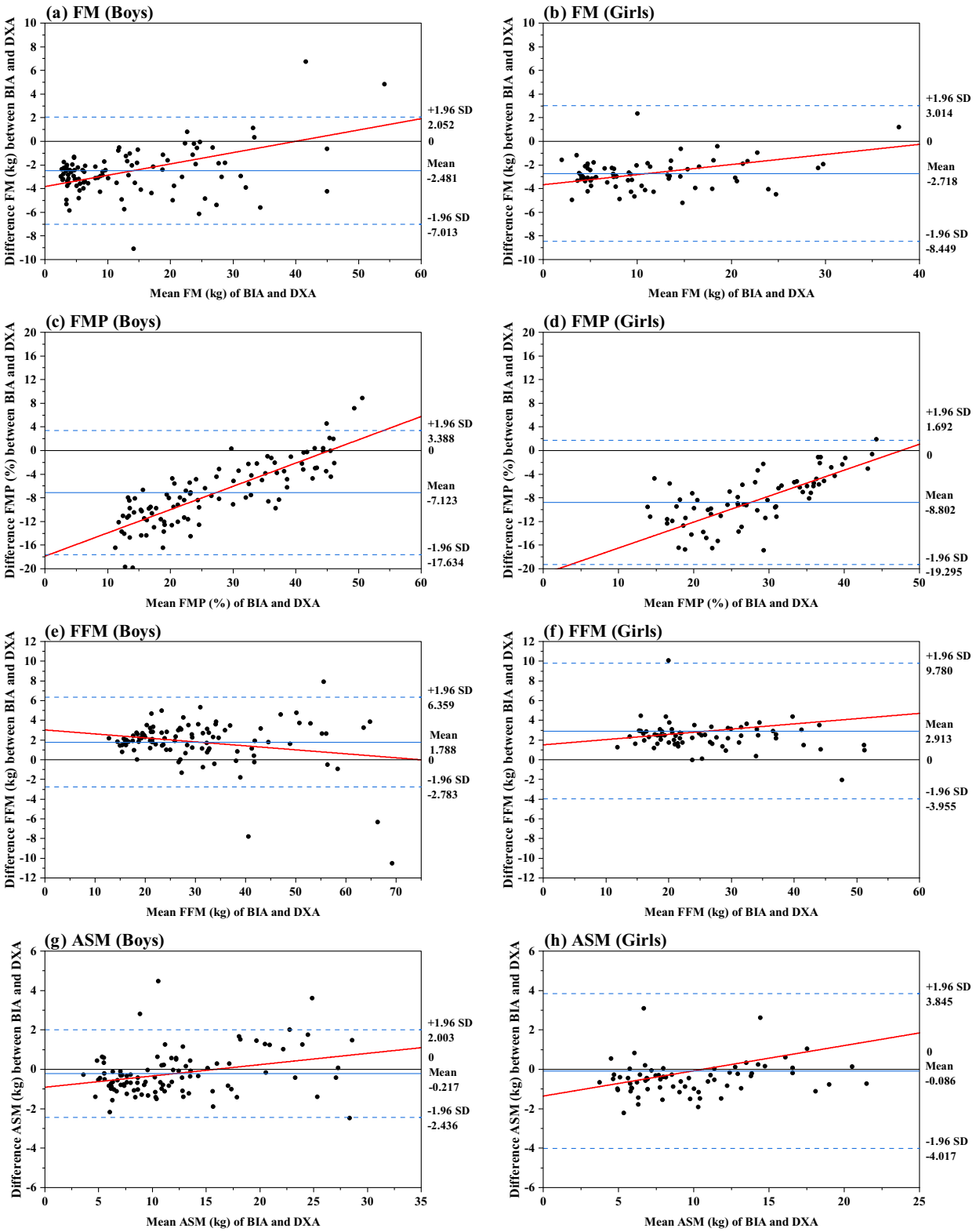

Fig. 2 Bland–Altman plots for body composition measurements assessed by bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA) among boys (a, c, e and g) and girls (b, d, f and h). *Y*-axis represents difference of body composition measurements (i.e. FM, FMP, FFM and ASM) measured by BIA minus those measured by DXA. *X*-axis represents mean of body composition measurements (i.e. FM, FMP, FFM and ASM) measured by BIA and DXA. Blue solid lines represent the mean difference between BIA and DXA. Blue dash lines represent the 95% limits of agreement. Red solid lines represent the linear regression between bias and means of measurements. ASM, appendicular skeletal muscle mass; FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage.

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Table 3 Comparison of body composition measured by bioelectrical impedance analysis and dual-energy X-ray absorptiometry between boys and girls

	Boys			Girls			
	ICC	Bias‡	MAPE	ICC	Bias‡	MAPE	*P* for difference†
FM (kg)							
Total	0.964	−2.5 ± 2.3**	31.1%	0.868	−2.7 ± 2.9**	32.6%	0.534
Age group							
Children	0.952	−2.5 ± 1.3**	32.4%	0.790	−2.8 ± 3.2 **	35.5%	0.489
Adolescents	0.959	−2.3 ± 4.8	25.2%	0.962	−2.4 ± 1.4 **	19.5%	0.950
P for difference§		0.738	0.221		0.670	0.010	
BMI category							
Underweight	0.604	−3.0 ± 0.9**	54.5%	0.127	−2.7 ± 0.9**	46.7%	0.345
Normal weight	0.582	−3.2 ± 1.4**	44.5%	0.804	−3.2 ± 0.8**	36.4%	0.497
Overweight	0.827	−3.1 ± 1.7**	16.8%	0.410	−2.6 ± 6.2	31.6%	0.870
Obesity	0.969	−0.9 ± 3.2	8.6%	0.961	−2.0 ± 1.3	12.7%	0.200
P for difference¶		<0.001	<0.001		<0.001	<0.001	
FMP (%)							
Total	0.750	−7.1 ± 5.4**	30.2%	0.519	−8.8 ± 5.4**	30.2%	0.123
Age group							
Children	0.733	−7.7 ± 4.8**	31.4%	0.467	−9.6 ± 5.4**	32.8%	0.031
Adolescents	0.812	−4.4 ± 6.9*	24.6%	0805	−5.1 ± 3.2**	18.7%	0.718
P for difference§		0.015	0.250		0.005	0.016	
BMI category							
Underweight	0.285	−11.1 ± 3.4**	53.9%	0.045	−11.2 ± 3.5**	46.0%	0.806
Normal weight	0.162	−10.4 ± 3.5**	43.4%	0.254	−10.4 ± 4.8**	35.4%	0.652
Overweight	0.485	−5.4 ± 2.8**	15.7%	0.198	−8.1 ± 6.8**	23.5%	0.149
Obesity	0.732	−1.6 ± 3.9*	7.9%	0.520	−4.3 ± 3.3**	11.8%	**0.025**
P for difference¶		<0.001	<0.001		<0.001	<0.001	
FFM (kg)							
Total	0.976	1.8 ± 2.3**	9.0%	0.895	2.9 ± 3.5**	13.5%	0.064
Age group							
Children	0.963	1.9 ± 1.3**	9.1%	0.839	3.2 ± 3.8**	14.9%	0.006
Adolescents	0.930	1.2 ± 4.8	8.8%	0.978	1.8 ± 1.4**	6.8%	0.691
P for difference §		0.233	0.849		0.188	0.080	
BMI category							
Underweight	0.914	2.6 ± 0.9**	12.4%	0.864	2.3 ± 0.7**	12.9%	0.325
Normal weight	0.961	2.6 ± 1.3**	11.5%	0.945	2.7 ± 0.7**	13.0%	0.179
Overweight	0.976	2.2 ± 1.4**	6.8%	0.711	5.4 ± 7.0*	22.2%	0.217
Obesity	0.976	0.1 ± 3.2	5.0%	0.985	1.3 ± 1.3*	6.1%	0.116
P for difference¶		<0.001	<0.001		0.002	0.025	
ASM (kg)							
Total	0.983	−0.2 ± 1.1	8.5%	0.899	−0.1 ± 2.0	12.9%	0.565
Age group							
Children	0.972	−0.5 ± 0.8**	8.5%	0.850	−0.1 ± 2.2	13.7%	0.111
Adolescents	0.946	0.9 ± 1.6*	8.1%	0.969	−0.3 ± 1.2	8.9%	0.035
P for difference§		<0.001	0.839		0.694	0.470	
BMI category							
Underweight	0.898	−0.4 ± 1.5	14.8%	0.871	−0.9 ± 0.6**	13.6%	0.389
Normal weight	0.978	−0.2 ± 1.1	8.6%	0.923	−0.1 ± 1.5	12.0%	0.607
Overweight	0.987	−0.5 ± 0.8*	5.3%	0.726	1.2 ± 3.6	20.4%	0.233
Obesity	0.985	−0.1 ± 1.2	6.7%	0.986	−0.5 ± 0.6*	6.1%	0.127
P for difference¶		0.522	0.002		0.025	0.333	

ASM, appendicular skeletal muscle mass; FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage; ICC, intraclass correlation coefficient; MAPE, mean absolute percentage error. * $P < 0.05$. ** $P < 0.001$. † To test the differences of bias between boys and girls. ‡ Bias was calculated by subtracting BIA-based body composition measurements (i.e. FM, FMP, FFM and ASM) from those measured by DXA. The paired *t* test was used to compare the means of bias between BIA and DXA. § To test the differences of bias or MAPE between children and adolescents. ¶ To test the differences of bias or MAPE among BMI category.

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1.9 kg, −17.5 to 4.7%, and − 3.0 to 5.6 kg in boys and −6.4 to 0.3 kg, −18.3 to 1.8%, and −0.8 to 5.5 kg in girls, respectively.

In addition, the absolute biases of BC measurements (i.e. FM, FMP and FFM) between the two methods in boys were significantly smaller than those in girls (all $P < 0.05$). When stratified by age group, the two methods' absolute biases of BC measurements had no statistical difference between children and adolescents, except for the absolute bias of FMP, which was significantly smaller in adolescents than in children. When stratified by BMI group, boys and girls with obesity had the lowest absolute bias or MAPE compared to the other three BMI categories. We investigated the relationship between the differences in BC measurements and BMI z-score to confirm the results from the categorical approach (Fig. S2). There were significant positive correlations between the differences in fat content and BMI z-score in boys ($\beta = 0.246$ in FM and 1.432 in FMP; all $P < 0.05$), whereas the difference in FFM between the two methods showed negative correlations with BMI z-score in boys ($\beta = -0.362$; $P = 0.004$). Similar patterns were also found in FMP estimation in girls ($\beta = 0.265$; $P = 0.456$).

Comparisons of BC measured by BIA and DXA

At the population level, there was excellent or good agreement between BIA and DXA for FM and FFM measurements (ICC: 0.964 and 0.976 in boys; ICC: 0.869 and 0.897 in girls), but the agreement of FMP measurements was moderate (ICC = 0.752 in boys and ICC = 0.522 in girls, Table 3). Body fat content obtained by BIA was significantly lower than by DXA (FM: bias = −2.5 kg in boys and bias = −2.7 kg in girls; FMP: bias = −7.1% in boys and bias = −8.8% in girls; all $P < 0.001$), but FFM obtained by BIA was significantly higher than by DXA (bias = 1.8 kg in boys and bias = 2.9 kg in girls, both $P < 0.001$, Fig. 2 and Table 3). At the individual level, the Bland–Altman plots showed the LOA for FM, FMP and FFM between the BIA and DXA were −7.0 to 2.1 kg, −17.6 to 3.4%, and −2.8 to 6.4 kg in boys and −8.4 to 3.0 kg, −19.3 to 1.7%, and −4.0 to 9.8 kg in girls, respectively (Fig. 2). In addition, BIA and DXA could obtain segmental BC measurements. There was excellent agreement between the two methods' estimates of ASM (ICC = 0.983 in boys and ICC = 0.900 in girls) with no significant differences between BIA and DXA. However, at the individual level, the LOA range of girls was larger than that of boys (−2.4 ∼ 2.0 kg in boys; −4.0 ∼ 3.8 kg in girls).

When stratified by age group, the two methods' absolute biases of FM and FFM had no statistical difference between children and adolescents, whereas the absolute bias of FMP was significantly smaller in children than in adolescents. When stratified by BMI group, boys and girls with obesity had the lowest absolute bias of BC measurements compared to the other three BMI categories, except for ASM, in which normal weight girls had the lowest absolute bias than the other three BMI groups. Regression analysis results between the differences and BMI z-score were shown in Figure S3. In boys, the gap between BIA and DXA in FM, FMP and FFM measurements became smaller as the boys' BMI category changed from underweight to obese. However, the difference in ASM did not show this trend relationship with BMI z-score. In girls, similar patterns were also found in FMP and ASM estimation, whereas the FM and FFM difference between the two methods has no significant trend relationship with the BMI z-score.

In addition, the MAPE of the body fat content measured by BIA and DXA was 30.2–32.6%, and the MAPE of FFM measured by BIA and DXA was 9.0% in boys and 13.5% in girls. The MAPE values showed a worse performance for BIA than ADP compared to DXA.

Comparisons of BC measured by BIA and ADP

At the population level, there was excellent or good agreement between BIA and ADP for fat content and FFM measurements (ICC: 0.917–0.980 in boys; ICC: 0.805–0.934 in girls, Table S1) with no statistical differences between the two methods, whereas the LOA at the individual level might not be clinically acceptable (Fig. S1).

Discussion

This study compared three commonly used BC measuring methods (ADP, BIA and DXA) among Chinese children and adolescents with BMI categories ranging from underweight to obesity. The results revealed that the agreements between paired methods were excellent or good at the population level. BIA is suitable for large-scale epidemiological field surveys because it is simple, cheap and easy to move. The high correlation between BIA and the other two methods provides a basis for further exploration of using BIA in field surveys. However, the precise estimates significantly varied across BMI categories, which should be considered when interpreting the measurements of paired methods. Moreover, the differences in paired methods might not be clinically acceptable at the individual level.

To the best of our knowledge, the study on the comparison of ADP and DXA in the paediatric population is limited, with only four studies in the past decade (Table S2). These four studies have reached the same conclusion: the ADP method correlated well with DXA, while ADP estimated lower fat mass content and higher FFM than DXA.[12,13,25,26] Our results confirmed and extended previous findings in a larger sample of Chinese children and adolescents. In addition, previous studies were mostly conducted in children with obesity, while the BMI categories of participants in our study ranged from underweight to obese. Therefore, our results additionally found that the difference in BC measurements between ADP and DXA was affected by BMI, and the absolute bias between the two methods tended to decrease with the increase of BMI, especially in boys. In other words, ADP and DXA showed better agreement and smaller absolute bias in children with obesity, consistent with our previous study in Chinese adults.[27]

Numerous comparative studies have evaluated the validity of BIA using DXA as the reference method in diverse populations (Table S3). Our results were consistent with most previous studies, with BIA estimating lower body fat content (FM and FMP) but greater FFM than DXA. However, some studies have reported inconsistent results. This discordance may be related to different parameters in different BIA devices used in each study.[28,29] These parameters included the way the current passes through the body (foot to foot, hand to foot or hand to hand), electrode numbers (two, four or eight), participant's position (standing or supine),

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the type of frequency (single-frequency or multi-frequency), and the in-built equations employed. According to the research comparing DXA and four commercially available BIA devices in Chinese children, there were statistical differences among the four brands of BIA equipment; even the biases were in the opposite direction.[25] Therefore, the comparative results of this study were only applicable to the specific BIA device, and caution should be taken when other BIA devices refer to this result.

Concerning ADP, the body weight is divided into FFM and FM, assuming that FFM density between individuals is constant.[18] BIA, also as a 2-C-based model, assumed that individuals have a constant specific resistivity, that is, the constant proportion of water in FFM.[30] The hydration variation will violate these two assumptions due to children and adolescents' growth and development. Studies have reported that hydration in children and adolescents is greater than in adults, resulting in lower FFM density and specific resistivity and an overestimation of FFM by ADP and BIA.[10,31] Besides, FFM hydration varies with weight status, with greater hydration in those affected by obesity than those not affected by obesity.[32] This variation explains the discrepancy between different BC measuring methods across BMI categories.

The present study has several strengths. First, to the best of our knowledge, this is the first comparative study of ADP and DXA in Chinese children and adolescents. Second, the study revealed that the discrepancies among the three commonly used BC measuring methods were different according to the BMI categories. However, this study also has some limitations. First, the three BC methods (ADP, BIA, and DXA) in this study are not the gold standard method; we could not identify which method is more accurate in measuring BC results among Chinese children and adolescents. Second, we used an age-predicted TGV in ADP estimation and did not consider sexual maturity. Third, the comparison results of BIA and DXA may only be applied to the Inbody 720 or the other multi-frequency octopolar BIA device with the same built-in formula. It is necessary to be cautious when using our findings to other BIA devices. Fourth, our results may only apply to children and adolescents; the difference in BC measurements between ADP and DXA in adults could refer to our earlier published articles.[27] Future comparative studies are required to carry out in infant and elder populations.

Conclusion

BC measurements obtained by the three commonly used methods were highly correlated in Chinese children and adolescents. However, ADP and BIA generally estimated a lower fat content but higher FFM than DXA. This difference was affected by BMI, and the absolute bias was the smallest in children with obesity. This study provided a basis for mutual reference of measurements between three widely used methods.

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Supporting Information

Additional Supporting Information may be found in the online version of this article at the publisher's web-site:

Figure S1. Bland–Altman plots for body composition measurements assessed by bioelectrical impedance analysis (BIA) and air displacement plethysmography (ADP) among boys (a, c and e) and girls (b, d and f). *Y*-axis represents difference of body composition measurements (i.e. FM, FMP and FFM) measured by BIA minus those measured by ADP. *X*-axis represents mean of body composition measurements (i.e. FM, FMP and FFM) measured by BIA and ADP. Blue solid lines represent the mean difference between measurements. Blue dash lines represent the 95% limits of agreement. Red solid lines represent the linear regression between bias and means of measurements. FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage.

Figure S2. Bland–Altman plots for body composition measurements assessed by air displacement plethysmography (ADP) and dual X-ray absorptiometry (DXA) across BMI-for-age z-score values among boys (a, c and e) and girls (b, d and f). *Y*-axis represents difference of body composition measurements (i.e. FM, FMP and FFM) measured by ADP minus those measured by DXA. Red solid lines represent the mean difference between ADP and DXA. Black dot dash lines represent the 95% limits of agreement. Blue solid lines and equations represent the linear regression between biases and BMI-for-age z-score values. FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage.

Figure S3. Bland–Altman plots for body composition measurements assessed by bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA) across BMI-for-age z-score values among boys (a, c, e and g) and girls (b, d, f and h). *Y*-axis represents difference of body composition measurements (i.e. FM, FMP, FFM and ASM) measured by BIA minus those measured by DXA. Red solid lines represent the mean difference between BIA and DXA. Black dot dash lines represent the 95% limits of agreement. Blue solid lines and equations represent the linear regression between biases and BMI-for-age z-score values. ASM, appendicular skeletal muscle mass; FFM, fat-free mass; FM, fat mass; FMP, fat mass percentage.

Table S1. Comparisons of body composition measured by bioelectrical impedance analysis and air displacement plethysmography among boys and girls

Table S2. Summary of comparative studies on body composition measurements between air displacement plethysmography (ADP) and dual-energy X-ray absorptiometry (DXA) among children and adolescents

Table S3. Summary of comparative studies on body composition measurements between bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA) among children and adolescents

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